

Answers

VARC

1.A	2.1423	3.D	4.A	5.C	6.B	7.A	8.A
9.3	10.B	11.C	12.A	13.C	14.3	15.B	16.C
17.D	18.C	19.D	20.C	21.C	22.B	23.D	24.2413

LRDI

25.A	26.B	27.A	28.D	29.4	30.4	31.A	32.B
33.3	34.6	35.C	36.A	37.A	38.60	39.40	40.84
41.D	42.45	43.35	44.50	45.D	46.9		

Quant

47.B	48.49	49.A	50.C	51.17	52.25	53.A	54.A
55.8	56.12	57.A	58.D	59.80	60.D	61.B	62.2
63.A	64.B	65.A	66.B	67.340	68.C		

Explanations

VARC

1. A

The paragraph explains why Denny focuses on “taste” instead of “judgment” and how this choice helps bring attention to an important but often ignored part of aesthetic and philosophical history. It first shows Denny pointing out a forgotten historical change, then explains that in the seventeenth and early eighteenth centuries, “taste” grew from a physical sense to a main idea in aesthetics and ethics. Finally, it discusses how tracing this history helps recover overlooked thinkers and ideas. The missing sentence fits best at blank (3) because, by this point in the paragraph, it has already been described how ‘taste’ expanded. The sentence adds to the discussion by explaining why taste is important: even though it was often seen as the same as judgment, it kept its link to pleasure, the body, and personal experience, which later theories often ignored. This directly builds on the previous point about taste’s central role in aesthetics and ethics.

Putting the sentence at blank (1) would be too early, since the paragraph has not yet talked about how the concept of ‘taste’ changed over time. Placing it at blank (2) would break up the flow from Denny’s motivation to the bigger historical story. If it went at blank (4), it would feel awkward because the paragraph there moves on to Denny’s method and the recovery of overlooked thinkers, not the difference between taste and judgment. So, blank (3) is the most coherent and logical position.

▶ VIDEO SOLUTION

2. 1423

Sentence 1 works best as the opening because it introduces the key idea of what “literature on screen” means and frames the debate by suggesting that adaptations might belong to both cinema and literature. Sentence 4 comes next because it explores this idea further and points out the main assumption in adaptation studies: that literature is the main way to understand adaptations. Sentence 2 then gives clear evidence of this bias by showing the contradiction that even though films now adapt material from many non-literary sources, academic studies still treat literature as cinema’s natural source. Finally, sentence 3 works as the conclusion because it explains *why* this contradiction continues. It argues that adaptation studies rely on literature’s cultural prestige to gain legitimacy, even if it means putting cinema in a secondary position. The most logical and clear order is therefore 1423.

VIDEO SOLUTION

3. D

The main idea of the passage is that the belief that humans always harm nature is not accurate. Throughout history, people have often created diverse landscapes that support many species. Research shows that human presence can increase biodiversity just as much as it can decrease it. The passage also questions the idea that nature only does well when people are not around.

Option D best sums up the main idea. It clearly contrasts the old belief that humans always harm nature and that nature does best without people, with the newer view that humans have often helped increase biodiversity. It also mentions the creation of different types of landscapes, matching the argument and balance of the passage.

The other options do not sum up the passage as completely as Option D. Option A is too narrow and focuses too much on people shaping nature on purpose, missing the bigger point about the human-versus-nature idea. Option B is too vague and does not include the history or the idea of mixed landscapes. Option C is closer, but it is too specific and only talks about certain times and actions, so it does not fully show the passage’s main message about how we view humans and nature.

VIDEO SOLUTION

4. A

The passage gives specific examples of AI-generated outputs to highlight a broader issue: the polite, *trustworthy tone* of AI systems can mask problems such as bias, distortion, and subtle influence on how users think. These examples show how those risks appear in real situations.

Seen under this light, **options B, C, and D** share a common feature. Each of these three options presents a clear example of an AI system producing an outcome that matches the author’s concern. This could be cultural bias, like changing Indian English to American English, representational bias, such as only showing male engineers and space explorers, or directional influence, like encouraging the author to describe a tech CEO more positively. These examples are used as evidence to show harm or distortion.

Option A falls outside this pattern. It does not show a problematic output, but instead gives a higher-level explanation from ChatGPT about its own ‘rhetorical’ design. Instead of showing bias or influence, it directly explains how trust can be built and possibly misused. So, **option A** does not fit with the rest, and is the odd one out.

VIDEO SOLUTION

5. C

The passage highlights a gap between what AI companies say their systems do, such as being neutral, inclusive, and trustworthy, and what these systems actually produce. We need to check each option to see if it supports this gap or not. For that, we review each option individually:

Option A clearly supports the gap. The system claims to be neutral or inclusive, but when asked for engineers and space explorers, it produces an “entirely male cast of characters.” This shows a clear difference between what is promised and what actually happens.

Option B also supports the gap. It points out the difference between the appearance of neutrality and the fact that this tone can make users accept biased or false answers. This sentence almost directly states the gap the passage discusses.

Option C does *not* support the gap. Here, the author holds back from making a judgment, saying they are “not aware of research” and can “only guess” why ChatGPT seemed to favour big tech or its CEO. Instead of showing a mismatch between claims and actions, this sentence adds uncertainty and does not reinforce the gap.

Option D does support the gap, but in a broader way. It shows that users benefit from AI products while also helping big tech gain more wealth and power. This highlights the difference between positive talk and real-world results.

So, the statement that does not support the gap is option C.

VIDEO SOLUTION



6. B

The phrase “a beige office building” is used to describe the *style and effect* of AI-generated language, not how AI explains or defends itself when questioned. The passage says people often call chatbot output “bland” or “generic,” which supports option A. It also mentions that OpenAI wants its products to “*sound like a colleague*,” using language that is “polite,” “empathetic,” “kind,” “rationally optimistic,” and “engaging,” which supports option D. The same design is said to help us feel safe and makes the product seem “professional” and “approachable,” which supports option C.

Option B is not connected to the “beige office building” metaphor. The part about ChatGPT blaming “biases in its training data” comes later and is about how the system answers questions about its mistakes, not about the bland, generic, and trust-building tone described by the metaphor. So, the correct answer is option B.

VIDEO SOLUTION

7. A

Let’s examine each option in relation to the passage.

Option A is the one the author is least likely to agree with. The author is concerned that AI’s neutral, friendly tone actually reduces critical thinking instead of encouraging it. She writes that the product’s “veneer of collegial neutrality could lull us into absorbing false or biased responses without much critical engagement.” This goes against the idea that neutrality helps critical thinking.

The author is cautious about option B but does not reject it. She says that ChatGPT “seemed to be guiding me to write a more positive book about big tech,” and even changed her description of Sam Altman. She also says she is “not aware of research” and can “only guess why it seemed that way.” This shows she is sceptical and careful, but she does not completely agree or disagree with the idea.

Option C matches the author’s view. Early on in the passage, she says that using big tech products makes users “complicit in big tech’s accumulation of wealth and power,” and calls us “both victims and beneficiaries.” This directly supports the claim; the author will definitely agree with this statement.

Option D also fits with the passage. The author says that funders “seek a return on their investment” and that it is easier to get people to use products like ChatGPT “if people see those products as trustworthy collaborators.” This connects AI’s neutral tone to business goals. The author will therefore agree with this statement as well.

So, the author is least likely to agree with option A.

VIDEO SOLUTION

8.A

The main argument of the passage is that Elizabeth Magie originally created the game as an anti-monopoly, anti-capitalist teaching tool, but her idea and message were later appropriated, simplified, and credited to Charles Darrow, leaving her largely unrecognised and poorly compensated. Option (a) captures this essence best because it highlights all the key elements together: (i) Magie’s left-wing feminist identity, (ii) the irony of her losing credit, (iii) the role of Darrow in selling a version of the game, (iv) the small payment she received, and (v) the denial of royalties. It reflects both the historical injustice and the irony that a game meant to critique monopolies ended up enacting one against its own creator.

Option (b) overstates Darrow and Parker Brothers as “celebrated icons” and focuses more on their success than on Magie’s original intent and the loss of her anti-capitalist message. Option (c) is incorrect because it repeats the very myth the passage is trying to debunk by portraying Darrow as the true innovator.

Option (d) shifts the focus too heavily onto Parker Brothers’ capitalist motives and underplays the central irony and misattribution surrounding Magie herself. Hence, option (a) is the best summary.

VIDEO SOLUTION

9.3

The paragraph is about Torres’s comedy show: its format, ideas, and delivery. Sentence 4 works as the opening because it introduces the show and explains its structure, making it the natural starting point. Sentence 1 continues this setup by shifting focus to Torres’ physical appearance. Sentence 5 then moves to his comic sensibility, developing the kind of ideas he explores, and Sentence 2 closes by characterising his delivery, tying back to how those ideas are presented. 4251, therefore, presents a coherent paragraph.

Sentence 3 breaks this flow because it introduces a specific “colour-scouting mission” and a personal audience reaction, which does not connect clearly to the descriptive sequence of the show’s structure, ideas, and style. Therefore, the correct order is **4-1-5-2**, and the odd sentence out is 3.

VIDEO SOLUTION

10.B

Let’s examine each option one by one.

Option A does not contradict the passage. The author describes astronomy as mostly “receptive” and biology as more “interventional,” but does not say these categories cannot overlap. The phrase “largely a ‘receptive’ science” suggests there could be some crossover. The passage also does not state that instruments cannot serve both roles, so this option matches the argument.

Option B does contradict the passage. The passage says mathematics may be “the last of the scientific disciplines to move into such technical embodiment,” meaning it has only recently become heavily dependent on instruments. However, this does not mean mathematics is “only now beginning to develop a scientific culture.” So, B misrepresents the claim.

Option C does not contradict the passage. The discussion about astronomy moving beyond the visible spectrum does not mean that microscopy cannot also go beyond visible limits. The passage does not say microscopy is limited to the visible spectrum or that it cannot move beyond it. So, option C does not contradict the argument; it just goes beyond what is stated.

Option D also does not contradict the passage. The passage mentions Newton improving telescopes and their use in astronomy, but it does not say *all* of Newton’s discoveries depended on instruments. Newton’s discovery of gravity without instruments does not conflict with the discussion about telescopes.

Therefore, only option B contradicts the passage.

VIDEO SOLUTION

CAT Percentile Predictor



11.C

We evaluate each option individually, against what the passage truly supports, and then reject it if it is an invalid inference.

Option A should be rejected. The passage does not set “interventionist instruments” and “embodied instruments” as separate categories. Instead, it states that sciences are “technologically, instrumentally embodied,” whether the instruments are receptive or interventional. The author’s point is about different *styles of embodiment*, not a contrast between embodiment and intervention. So, this inference misunderstands the main idea.

Option B should also be rejected. The passage credits Galileo with observing “the phases of Venus” and “the satellites of Jupiter,” saying “it was with his own telescopes that he made the observations launching early modern astronomy.” Newton’s reflecting telescopes are mentioned as *later improvements*, not as the tools for those discoveries. This inference does not match the historical order described.

Option C is valid and should NOT be rejected. The passage clearly compares astronomy and biology based on how they use instruments. Astronomy is called “largely a ‘receptive’ science, dependent upon instrumentation which can detect and receive emissions,” while biology, especially with biotechnology, uses instruments that are “far more interventional in style.” This comparison is a key part of the author’s point.

Option D should be rejected. The passage links the “new astronomy” to the use of the full electromagnetic spectrum, starting with radio astronomy in the twentieth century. Newton’s reflecting telescopes belong to optical astronomy and the trend of increasing magnification and resolution. Calling Newton’s work the start of the “new astronomy” does not fit the passage.

Therefore, option C is the only valid inference.

VIDEO SOLUTION

12. A

The passage describes instruments as active, technology-based tools that play a role in practice. They help us see more, take action, or support skilled work. Whether they receive information, like telescopes, or allow intervention, like biotechnological tools, these instruments are not passive. They influence what we can notice, do, or understand. Looking into the options, we can say that the only option that doesn't fit this common definition of being an 'active mediator' is a milestone (Option A). A kitchen oven (Option B) changes materials by controlling conditions, a scalpel (Option C) acts directly on biological matter, and a saxophone (Option D) needs skill to create music. A milestone is just a fixed marker that shows distance. It does not help us perceive, change, or take part in any technical process as described in the passage. It does not detect or affect anything. So, the characterisation is least applicable to option A.

VIDEO SOLUTION

13. C

The quoted line says that the sciences "are today highly instrumentalised and could not progress successfully without constant improvements upon the respective instrumental trajectories." This means scientific progress is directly connected to better instruments. Option C matches this idea well. The passage talks about astronomy and microbiology separately, showing that better telescopes helped astronomy grow and that tools like gene-splicing shape modern biology. It then says both fields rely on ongoing improvements in their instruments. These examples show that in both fields, scientific progress has followed improvements in instruments.

The other options fail: Option A changes the idea from dependence to "must be respected," which adds a value judgment or obligation not found in the passage. Option B talks about the "growth of scientific technologies" but misses the clear link between progress and better instruments. Option D mentions "scientific constants," which the passage does not discuss. So, the correct answer is **Option C**.

VIDEO SOLUTION

14. 3

The sentences discuss new scientific research on PFAS. They cover what these chemicals are, what the research found about their health effects, and why these findings are important for medicine. The main focus is on the research, its implications, and how it may help identify diseases in the future.

Sentences 2, 1, 4, and 5 follow a logical order. Sentence 2 introduces the new research and its main claim about gene activity and health problems. Sentence 1 gives background on what PFAs are and why they are used. Sentence 4 builds on the research by suggesting PFAS could be linked to more diseases that have not yet been identified. Sentence 5 wraps up by explaining why these findings matter, showing how they help us understand diseases and improve early detection and treatment.

Sentence 3 does not fit well with the others. It is accurate, but it explains why PFAS are called "forever chemicals" because they stay in the environment. None of the other sentences discusses this idea; moreover, statement 3 does not add to the main points about gene activity, disease, or medical impact. So, it feels like extra information that is not directly related. Therefore, the odd sentence out is **3**.

VIDEO SOLUTION

15. B

The Sentence mentions "the region's Western customers," implying that the region in question is not in the West. It follows that the area being mentioned must be Dhaka and the surrounding places where the Muslin was produced and sold. The sentence before Blank-4 mentions varied places, and therefore, the Sentence cannot be placed in Blank-4. Starting from "Handspun from a rare cotton...", until "... a 60' length fit inside a snuffbox," the passage discusses how the Dhaka Muslin was made. Placing the Sentence in between would disrupt the flow of the paragraph and the surrounding context. Blank-1 is the most suitable choice, since the Sentence can then follow the preceding mentions of "silty banks of Meghna river" and Dhaka itself. Option B is the correct answer.

VIDEO SOLUTION



16. C

The passage describes how Mexican tetra cavefish have developed specific traits that help them survive in caves. These include losing their eyes, changing their metabolism, and controlling inflammation to cope with limited and unpredictable food. Of the choices, we need to identify which statement does not describe an adaptation in cavefish, and for that, we evaluate each option individually.

Option A [... *maintain expensive tissues they aren't using* ...] shows an example of adaptation. The passage clearly says that keeping unused tissues like eyes is a disadvantage in caves where energy is limited. Losing eyes is described as an evolutionary change that helps save energy.

Option B [... *fish are likely surviving on a meager diet* ...] talks about the environmental pressure that cavefish have adapted to. The limited diet of bat feces and organic waste shows *why* adaptations for absorbing and storing nutrients are helpful. Even though this sentence describes the environment, it is still part of the adaptation story because it explains what drives these traits.

Option C [... *cells get too big, they can burst* ...] does not describe an adaptation in cavefish. Instead, it explains why fat storage causes inflammation in humans and other animals. This sentence gives background information, but the passage later points out that cavefish *do not* have this inflammation. So, this statement is about other organisms, not about cavefish adaptation.

Option D [... *cavefish can sometimes get very fat* ...] also describes an adaptation. Cavefish can become very fat even when food is scarce, which shows their evolved ability to overeat when possible and store nutrients well. This is an advantage in an environment where food is unpredictable.

So, the statement that does not describe an adaptation in Mexican tetra cavefish is option C.

VIDEO SOLUTION

17. D

The idea of "apparent inefficiency is unavoidable" refers to the fact that cavefish embryos begin developing eyes only for those eye cells to die off later, and this process is, however, unavoidable. The explanation must therefore point to a constraint in early development that makes it impossible to avoid initiating eye formation in the first place. The correct option must directly account for *why this sequence cannot be skipped*.

Option D gives this explanation. It says that Mexican tetra cavefish develop in the same way as their non-cave-dwelling relatives at first. The passage also says that eye and brain development are "completely intertwined" early on. Since cavefish embryos follow the same path as non-cave-dwelling fish, evolution cannot remove eye formation at the start without affecting brain development. Hence, starting eye formation and then dismantling it becomes unavoidable.

The other options do not work because they focus on selective pressure or later results, not on why eye formation must start. Option A talks about why eyes are costly for adult fish, not why embryos begin forming them. Option B incorrectly blames eye loss on darkness instead of on the intrinsic developmental processes. Option C mentions changes that happen after eye loss, but does not explain why the inefficient developmental sequence itself cannot be avoided.

[VIDEO SOLUTION](#)

18. C

The experiment tests Riddle's idea that carotenoid accumulation (yellow fat) is not linked to eye loss but serves another purpose, like reducing inflammation. To disprove this, results must show a direct connection between losing eyes and carotenoid accumulation.

Option C would directly invalidate this inference. If *only* the eyeless offspring had yellow fat, it would provide a strong link between the absence of eyes and carotenoid buildup. This would go against Riddle's statement that there is "no link between eye presence or size and the accumulation of carotenoids," and would support the earlier idea that carotenoids build up because cavefish lack eyes.

The other options do not challenge her idea. Options A and D show that white fat can appear whether or not the fish have eyes. Option B shows that fish with eyes can also have yellow fat. These results fit with Riddle's claim that carotenoid buildup does not depend on eye presence. So, only option C would disprove her inference.

[VIDEO SOLUTION](#)

19. D

The passage explicitly states Riddle's conclusion: "*Instead, Riddle thinks these carotenoids may be another adaptation to suppress inflammation.*" This directly points to carotenoids playing a role in controlling inflammation, especially in the context of fat cells becoming large and potentially causing inflammatory damage. Therefore, the correct answer is **option D**.

The other options are either not mentioned in the passage or, if mentioned, are discussed in a different context. Option A is mentioned only as an initial hypothesis that carotenoids might be linked to eye loss, but this idea is explicitly rejected after cross-breeding shows "*no link between eye presence or size and the accumulation of carotenoids.*" Option B is not mentioned at all in connection with carotenoids; nutrient storage is discussed as a general metabolic adaptation, not as a function of carotenoids. Option C refers merely to the observed yellow colour of fat, which the passage treats as a visible effect, not as the biological purpose of carotenoids.

[VIDEO SOLUTION](#)

20. C

The sentence "The pursuit of such a topic speaks to the valuable knowledge produced from bridging disciplines and combining material from both the arts and the sciences to better understand the human condition" refers to the importance of "bridging" or intermixing of the material from both arts and sciences in understanding the human condition. The statement that best captures this is provided in option C.

Option A is incorrect as "bridging of human condition" is a wrong inference. Option B, although partly correct, misses the message of "bridging" which forms the core of the sentence. Option D is incorrect as it reverses the relationship of the importance of knowledge from arts and science in our understanding of the human condition by writing that human emotions are essential contributors to our understanding of arts and sciences.

[VIDEO SOLUTION](#)



21. C

The second paragraph primarily emphasises the importance of literature in providing insights into the emotional dimension of physical and cultural geography, which is all too often missing from scientific records. The paragraph notes that this emotional or spiritual component is primarily subjective and that the history of a place also exists in memory and imagination.

Accordingly, options A, B, and D follow. Option C is extreme in that it says that the emotional and spiritual experience of a place can replace a surveyor's map, and is incorrect. Although the emotional and spiritual experience can complement the factual information about geographic and physical world provided by the scientific record, it cannot replace it.

[VIDEO SOLUTION](#)

22. B

The paragraph states that "For example, a literary description of place may involve not only the environment and geography but the noises and quality of light, or how people from different races or classes can experience the same place in different ways linked to those racial or class disparities."

Based on this and the information provided in the passage, it would be wrong to consider the fact that scientific approaches to place are more (or less) accurate than literary ones. The most we can say is that both are useful and complementary. Option A is incorrect.

Option B follows from the example of literary description, which examines how people from different backgrounds experience the same place differently, and is therefore the correct answer.

Option C mentions architects and the diverse methods they use, which is not the reason why the passage provides the example of literary descriptions. Similarly, option D examines in detail how the absence of people and animals affects the inhabitants and fails to capture the broader message the passage seeks to convey through the example provided.

Option B is the correct answer.

[VIDEO SOLUTION](#)

23. D

The passage asserts that "Histories of places live on in many forms, one of which is the human memory or imagination." The passage also implies that these places and their "emotional component, although subjective, is no less 'real' than a surveyor's map." It follows that, in literature, descriptions of places do not need satellite imagery or other visual aids to give a "real" sense of the place. If this were to be false, the arguments made in the passage would be contradicted. Option A is not the correct answer.

From "Both real and imaginary landscapes provide insight into the human experience of place", we can also infer that if option B were to be false, the information would contradict the passage. Option B is not the correct answer.

From "Literature can bring together material that spans different branches of science. For example, a literary description of place may involve not only the environment and geography but the noises and quality of light, or how people from different races or classes can experience the same place in different ways linked to those racial or class disparities. Literary texts can also account for the way in which absence—of other people, animals, and so on—affects a human observer or inhabitant", we can infer that literature can be a powerful device that can produce vivid realities through descriptions. If option C were to be false, it would contradict the information provided in the passage. Option C is not the correct answer.

From "In human experience, geographical places have a spiritual or emotional component in addition to and as part of a physical layout and topography. This emotional component, although subjective, is no less "real" than a surveyor's map," we can infer that the physical and emotional experience of a place is subjective. Therefore, if option D were to be false, the arguments provided in the passage will not be contradicted. Option D is the correct answer.

VIDEO SOLUTION

24. 2413

Sentence 1 mentions "compounds" while sentence 3 talks about them in detail. Sentence 3 starts with "these compounds" making sentence 1 and 3 an adjacent pair in that order. Similarly, sentence 2 introduces "a scent" that is elaborated upon in sentence 4, implying that sentences 2 and 4 are a pair in that order. The "slow chemical changes" mentioned in sentence 4 are elaborated further in sentences 1 and 3. The most suitable sequence, therefore, would be 2413.

VIDEO SOLUTION

LRDI

25. A

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Ananya and Bhaskar were Gurubhai from 2019 to 2020. Since 2020 is the only option present during this period, it has to be the answer.

Hence, the correct answer is option A.

[VIDEO SOLUTION](#)



26. B

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

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2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Charu began her training under Pandit Meghnath in 2015.

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)

27. A

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Bhaskar and Devendra were Gurubhai from 2021 to 2022. Since 2022 is the only option present during this period, it has to be the answer.

Hence, the correct answer is option A.

[VIDEO SOLUTION](#)

28. D

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Statement 1) Charu was training under Ustad Samiran in 2018. This is false as she was training under Pandit Meghnath in 2018.

Statement 2) Ananya was training under Ustad Samiran in 2015. This is false as she was training under Pandit Meghnath in 2015.

Statement 3) Ananya was training under Ustad Samiran in 2018. This is false as she was not training in 2018.

Statement 4) Charu was training under Ustad Samiran in 2019. This is true.

Hence, the correct answer is option D.

VIDEO SOLUTION

29.4

Representing Ananya Raga, Bhaskar Tala, Charu Veena, and Devendra Sur as A, B, C and D, respectively, for easy usage. Also using PM, US, and AR for Pandit Meghnath, Ustad Samiran, and Acharya Raghunath, respectively, for easy reference.

We are given that AR did not train during the periods of 2015-2018 and 2021-2024.

Other information provided is that PM, US, and AR had a span of 2, 3, and 4 consecutive years for each student, in some order. We also know that all the gurus trained each of the students for exactly one span.

In clue 4, we are given that A started the training in 2013 under PM, and B started training in 2013 under US.

Putting all the known information in the table, we get,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	
2014			
2015			
2016			
2017			
2018			
2019			
2020			
2021			
2022			
2023			
2024			

Now, if we look at the table, we can see that AR is training for only 4 years, with two years each, separated by a few years. Therefore, the span of AR must be 2 years because we know that he is not teaching continuously for 3 or 4 years.

Since we know that each guru taught all the students, and AR only has 4 years to teach all the students, he must definitely teach 2 students at a time each year, as we are told that no guru taught more than 2 students in a single year. We also know that A and B were taught by PM and US in 2013, respectively. Therefore, we can conclude that C and D were taught by AR during both 2013 and 2014, as AR's span is 2 years.

We can also conclude that A and B were taught by AR during both 2019 and 2020, as AR's span is 2 years, and they were the only students left for AR's class.

We are also given in clue 1 that US did not train more than 1 student in any year, and there are a total of 12 years from 2013 to 2024. If the span of US is 4 years, then he would need a total of 16 years to teach every student, as he is not teaching more than 1 student in any year, but he only taught for 12 years, so the only possibility is for US to have a span of 3 years and PM to have a span of 4 years. We can also conclude that US taught B from 2013 to 2015, as he has a span of 3 years.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014		B	C,D
2015		B	
2016			
2017			
2018			
2019			A,B
2020			A,B
2021			
2022			
2023			
2024			

We are also given that A, D and B, C were never Gurubhai, and all the other pairs were Gurubhai for exactly 2 years. Therefore, there must be two years each of {A, B}, {A, C}, {B, D}, and {C, D} with classes together. We have already completed the 2 years of {A, B} and {C, D}, so the ones left are two years of {A, C} and {B, D}. We know that the US did not take classes of 2 students together, and we have already filled the schedule of AR, so the only one left is PM, and he must have the other two left.

Since the span of PM is 4, he taught A from 2013 to 2016 and during this span, there must be 2 years of {A, C}, as we know that A will not be taught again after 2016. To have an exact span of 2 years together, C must start in 2015, as it is the only possibility, and he will teach C until 2018, as he has a span of 4 years.

There must be 2 years of {B, D} in the remaining time period of PM. In 2019, A and B were already enrolled with AR, and C had already completed his time with PM. So, the only person possible to have class in 2019 with PM is D, and D will have classes with PM until 2022. Now for 2 years of {B, D}, D must have his classes start from 2021, as it is the only possibility.

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C		
2017	C		
2018	C		
2019	D		A,B
2020	D		A,B
2021	D,B		
2022	D,B		
2023	B		
2024	B		

There cannot be any gap in the case of US, and in 2016, the only possibility is D as A, C were having classes with PM, and B already had his classes with US. D had classes from 2016 to 2018. In 2019, A, B and D were already having classes, so the only possibility is for C to have classes with US from 2019 to 2021, and finally A will have his classes from 2022 to 2024 with US, as he is the only person left to have classes with US.

The final table looks like,

YEAR	Pandit Meghnath(PM)	Ustad Samiran(US)	Acharya Raghunath(AR)
2013	A	B	C,D
2014	A	B	C,D
2015	A,C	B	
2016	A,C	D	
2017	C	D	
2018	C	D	
2019	D	C	A,B
2020	D	C	A,B
2021	D,B	C	
2022	D,B	A	
2023	B	A	
2024	B	A	

Between 2013 and 2024, the years 2017, 2018, 2023, and 2024 were the only years when only two of these four musicians were training under these three Gurus.

Hence, the correct answer is 4.

[VIDEO SOLUTION](#)

30.4

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

If all the four-author-type books are counted for all four authors separately, then we get the sum to be $4 * 2 = 8$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Putting the values in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

We are given that each author has at least one book of each type. In the case of Arman, there are three books left to be assigned, and there are 3 types of books left, so each of them has to be equal to 1 for the above condition to be satisfied.

In the case of Brajen, we are told that he had an equal number of single-author and two-author books.

Case 1: If the equal number is 1, then the three-author type books become 4, and we are given that both Chintan and Devon had more three-author books than Brajen, which is not possible in this case, as both of them combined will be left with 5 books together and whatever we split 5 books there wont be a case of both of them having more books than Brajen.

Case 2: If the equal number is 2, then the three-author type books become 2, and we are given that both Chintan and Devon had more three-author books than Brajen, which is possible in this case, as both of them combined will be left with 6 books together and the only way to split 6 books such that both of them are greater than 2 is to split them into 3, 3 and it is the only possible way.

Case 3: If the equal number is greater than or equal to 3, then the three-author type books become 0 or negative, which can directly be eliminated, as we are given that all authors wrote at least 1 book of each type.

So, we can eliminate both cases 1 and 3, and we will be left with the case of Brajen having 2 books each of Single-author, Two-author, and Three-author types. Chintan and Devon have three books each of the three-author type.

Filling up the table with these values, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

We are given that each author has a different number of single-author type books. The only possibilities for the single-author books of Chintan and Devon are 3 and 4 in some order.

If the single-author books of Chintan are 3, then the double-author books of Chintan become 4, the single-author books of Devon become 4, and the double-author books of Devon become 1.

If the single-author books of Chintan are 4, then the double-author books of Chintan become 3, the single-author books of Devon become 3, and the double-author books of Devon become 2.

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

The total number of two-author and three-author papers written by Brajen = $2 + 2 = 4$.

Hence, the correct answer is 4.

[VIDEO SOLUTION](#)



31. A

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

If all the four-author-type books are counted for all four authors separately, then we get the sum to be $4 * 2 = 8$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Putting the values in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

We are given that each author has at least one book of each type. In the case of Arman, there are three books left to be assigned, and there are 3 types of books left, so each of them has to be equal to 1 for the above condition to be satisfied.

In the case of Brajen, we are told that he had an equal number of single-author and two-author books.

Case 1: If the equal number is 1, then the three-author type books become 4, and we are given that both Chintan and Devon had more three-author books than Brajen, which is not possible in this case, as both of them combined will be left with 5 books together and whatever we split 5 books there wont be a case of both of them having more books than Brajen.

Case 2: If the equal number is 2, then the three-author type books become 2, and we are given that both Chintan and Devon had more three-author books than Brajen, which is possible in this case, as both of them combined will be left with 6 books together and the only way to split 6 books such that both of them are greater than 2 is to split them into 3, 3 and it is the only possible way.

Case 3: If the equal number is greater than or equal to 3, then the three-author type books become 0 or negative, which can directly be eliminated, as we are given that all authors wrote at least 1 book of each type.

So, we can eliminate both cases 1 and 3, and we will be left with the case of Brajen having 2 books each of Single-author, Two-author, and Three-author types. Chintan and Devon have three books each of the three-author type.

Filling up the table with these values, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

We are given that each author has a different number of single-author type books. The only possibilities for the single-author books of Chintan and Devon are 3 and 4 in some order.

If the single-author books of Chintan are 3, then the double-author books of Chintan become 4, the single-author books of Devon become 4, and the double-author books of Devon become 1.

If the single-author books of Chintan are 4, then the double-author books of Chintan become 3, the single-author books of Devon become 3, and the double-author books of Devon become 2.

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

Statement 1: Chintan wrote exactly three two-author papers.

This is not necessarily true, as he could have written four books as well.

Statement 2: Chintan wrote more single-author papers than Devon.

This is not necessarily true, as the opposite can be possible.

So, neither 1 nor 2 is necessarily true.

Hence, the correct answer is option A.

 VIDEO SOLUTION

32. B

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

If all the four-author-type books are counted for all four authors separately, then we get the sum to be $4 * 2 = 8$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Putting the values in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

We are given that each author has at least one book of each type. In the case of Arman, there are three books left to be assigned, and there are 3 types of books left, so each of them has to be equal to 1 for the above condition to be satisfied.

In the case of Brajen, we are told that he had an equal number of single-author and two-author books.

Case 1: If the equal number is 1, then the three-author type books become 4, and we are given that both Chintan and Devon had more three-author books than Brajen, which is not possible in this case, as both of them combined will be left with 5 books together and whatever we split 5 books there won't be a case of both of them having more books than Brajen.

Case 2: If the equal number is 2, then the three-author type books become 2, and we are given that both Chintan and Devon had more three-author books than Brajen, which is possible in this case, as both of them combined will be left with 6 books together and the only way to split 6 books such that both of them are greater than 2 is to split them into 3, 3 and it is the only possible way.

Case 3: If the equal number is greater than or equal to 3, then the three-author type books become 0 or negative, which can directly be eliminated, as we are given that all authors wrote at least 1 book of each type.

So, we can eliminate both cases 1 and 3, and we will be left with the case of Brajen having 2 books each of Single-author, Two-author, and Three-author types. Chintan and Devon have three books each of the three-author type.

Filling up the table with these values, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

We are given that each author has a different number of single-author type books. The only possibilities for the single-author books of Chintan and Devon are 3 and 4 in some order.

If the single-author books of Chintan are 3, then the double-author books of Chintan become 4, the single-author books of Devon become 4, and the double-author books of Devon become 1.

If the single-author books of Chintan are 4, then the double-author books of Chintan become 3, the single-author books of Devon become 3, and the double-author books of Devon become 2.

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

Statement 1: Arman wrote three-author papers only with Chintan and Devon.

There are in total 3 three-author books, and Chintan and Devon are authors of all three of them according to the table. So, Arman wrote the three-author papers only with Chintan and Devon. So, the statement is necessarily true

Statement 2: Brajen wrote three-author papers only with Chintan and Devon.

There are in total 3 three-author books, and Chintan and Devon are authors of all three of them according to the table. So, Brajen wrote the three-author papers only with Chintan and Devon. So, the statement is necessarily true.

So, both I and 2 are necessarily true.

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)

33.3

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

If all the four-author-type books are counted for all four authors separately, then we get the sum to be $4 * 2 = 8$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Putting the values in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

We are given that each author has at least one book of each type. In the case of Arman, there are three books left to be assigned, and there are 3 types of books left, so each of them has to be equal to 1 for the above condition to be satisfied.

In the case of Brajen, we are told that he had an equal number of single-author and two-author books.

Case 1: If the equal number is 1, then the three-author type books become 4, and we are given that both Chintan and Devon had more three-author books than Brajen, which is not possible in this case, as both of them combined will be left with 5 books together and whatever we split 5 books there wont be a case of both of them having more books than Brajen.

Case 2: If the equal number is 2, then the three-author type books become 2, and we are given that both Chintan and Devon had more three-author books than Brajen, which is possible in this case, as both of them combined will be left with 6 books together and the only way to split 6 books such that both of them are greater than 2 is to split them into 3, 3 and it is the only possible way.

Case 3: If the equal number is greater than or equal to 3, then the three-author type books become 0 or negative, which can directly be eliminated, as we are given that all authors wrote at least 1 book of each type.

So, we can eliminate both cases 1 and 3, and we will be left with the case of Brajen having 2 books each of Single-author, Two-author, and Three-author types. Chintan and Devon have three books each of the three-author type.

Filling up the table with these values, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

We are given that each author has a different number of single-author type books. The only possibilities for the single-author books of Chintan and Devon are 3 and 4 in some order.

If the single-author books of Chintan are 3, then the double-author books of Chintan become 4, the single-author books of Devon become 4, and the double-author books of Devon become 1.

If the single-author books of Chintan are 4, then the double-author books of Chintan become 3, the single-author books of Devon become 3, and the double-author books of Devon become 2.

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

If Devon wrote more than one two-author paper, he would have written 2 papers, making the number of two-author papers written by Chintan 3.

Hence, the correct answer is 3.

[VIDEO SOLUTION](#)

34.6

Let us try to arrange the hoops and the spheres in increasing order of their diameter according to the information given,

In clue 1, we are given that B1 and B6 made a ping on H4, and B5 did not.

So, we can definitely say that,

$(B1, B6) < H4 < B5 \rightarrow (1)$

In clue 2, we are given that B4 made a ping on H3, but B1 did not.

So, we can definitely say that,

$$B4 < H3 < B1 \text{ --(2)}$$

In clue 3, we are given that all balls, except B3, made pings on H1.

So, we can definitely say that,

$$(B1, B2, B4, B5, B6) < H1 < B3 \text{ --(3)}$$

In clue 4, we are given that none of the balls, except B2, made a ping on H2.

So, we can definitely say that,

$$B2 < H2 < (B1, B3, B4, B5, B6) \text{ --(4)}$$

Combining (1) and (2), we can definitely say that

$$B4 < H3 < B1 < H4 < B5 \text{ --(5)}$$

The only ball that we do not have enough information is about B6, as it can be either less than H3 or greater than H3 and less than H4.

Combining (3), (4) and (5), we get,

$$B2 < H2 < B4 < H3 < B1 < H4 < B5 < H1 < B3$$

The sphere B6 can be placed in 2 positions, giving us two possible inequalities, which are

$$B2 < H2 < (B4, B6) < H3 < B1 < H4 < B5 < H1 < B3$$

$$B2 < H2 < B4 < H3 < (B1, B6) < H4 < B5 < H1 < B3$$

The position of B6 is not clear, and except for that, all the positions are fixed.

Pings made by B1 = H4, H1 = 2

Pings made by B2 = H2, H3, H4, H1 = 4

Pings made by B3 = 0

Total number of pings by B1, B2 and B3 = 2 + 4 + 0 = 6.

Hence, the correct answer is 6.

 VIDEO SOLUTION

35. C

Let us try to arrange the hoops and the spheres in increasing order of their diameter according to the information given,

In clue 1, we are given that B1 and B6 made a ping on H4, and B5 did not.

So, we can definitely say that,

$$(B1, B6) < H4 < B5 \text{ --(1)}$$

In clue 2, we are given that B4 made a ping on H3, but B1 did not.

So, we can definitely say that,

$$B4 < H3 < B1 \text{ --(2)}$$

In clue 3, we are given that all balls, except B3, made pings on H1.

So, we can definitely say that,

$$(B1, B2, B4, B5, B6) < H1 < B3 \text{ --(3)}$$

In clue 4, we are given that none of the balls, except B2, made a ping on H2.

So, we can definitely say that,

$$B2 < H2 < (B1, B3, B4, B5, B6) \text{ --(4)}$$

Combining (1) and (2), we can definitely say that

$$B4 < H3 < B1 < H4 < B5 \text{ --(5)}$$

The only ball for which we do not have enough information is about B6, as it can be either less than H3 or greater than H3 and less than H4.

Combining (3), (4) and (5), we get,

$$B2 < H2 < B4 < H3 < B1 < H4 < B5 < H1 < B3$$

The sphere B6 can be placed in 2 positions, giving us two possible inequalities, which are

$$B2 < H2 < (B4, B6) < H3 < B1 < H4 < B5 < H1 < B3$$

$$B2 < H2 < B4 < H3 < (B1, B6) < H4 < B5 < H1 < B3$$

The position of B6 is unclear, and except for that, all the other positions are fixed.

Option A) $B4 < B5 < B3$. This is definitely true.

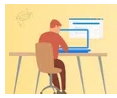
Option B) $B2 < B1 < B5$. This is definitely true.

Option C) $B1 < B6 < B3$. This need not be true, as we do not know which of B1 and B6 has the larger diameter.

Option D) $B1 < B5 < B3$. This is definitely true.

Hence, the correct answer is option C.

 VIDEO SOLUTION



FREE

CAT Sectional Tests

**36. A**

Let us try to arrange the hoops and the spheres in increasing order of their diameter according to the information given,

In clue 1, we are given that B1 and B6 made a ping on H4, and B5 did not.

So, we can definitely say that,

$$(B1, B6) < H4 < B5 \text{ --(1)}$$

In clue 2, we are given that B4 made a ping on H3, but B1 did not.

So, we can definitely say that,

$$B4 < H3 < B1 \text{ --(2)}$$

In clue 3, we are given that all balls, except B3, made pings on H1.

So, we can definitely say that,

$$(B1, B2, B4, B5, B6) < H1 < B3 \text{ --(3)}$$

In clue 4, we are given that none of the balls, except B2, made a ping on H2.

So, we can definitely say that,

$$B2 < H2 < (B1, B3, B4, B5, B6) \text{ --(4)}$$

Combining (1) and (2), we can definitely say that

$$B4 < H3 < B1 < H4 < B5 \text{ --(5)}$$

The only ball for which we do not have enough information is about B6, as it can be either less than H3 or greater than H3 and less than H4.

Combining (3), (4) and (5), we get,

$$B2 < H2 < B4 < H3 < B1 < H4 < B5 < H1 < B3$$

The sphere B6 can be placed in 2 positions, giving us two possible inequalities, which are

$$B2 < H2 < (B4, B6) < H3 < B1 < H4 < B5 < H1 < B3$$

$$B2 < H2 < B4 < H3 < (B1, B6) < H4 < B5 < H1 < B3$$

The position of B6 is unclear, and except for that, all the other positions are fixed.

We know the correct order of the diameters of the hoops is $H2 < H3 < H4 < H1$.

Hence, the correct answer is option A.

[VIDEO SOLUTION](#)**37. A**

Let us try to arrange the hoops and the spheres in increasing order of their diameter according to the information given,

In clue 1, we are given that B1 and B6 made a ping on H4, and B5 did not.

So, we can definitely say that,

$$(B1, B6) < H4 < B5 \text{ --(1)}$$

In clue 2, we are given that B4 made a ping on H3, but B1 did not.

So, we can definitely say that,

$$B4 < H3 < B1 \text{ --(2)}$$

In clue 3, we are given that all balls, except B3, made pings on H1.

So, we can definitely say that,

$$(B1, B2, B4, B5, B6) < H1 < B3 \text{ --(3)}$$

In clue 4, we are given that none of the balls, except B2, made a ping on H2.

So, we can definitely say that,

$$B2 < H2 < (B1, B3, B4, B5, B6) \text{ --(4)}$$

Combining (1) and (2), we can definitely say that

$$B4 < H3 < B1 < H4 < B5 \text{ --(5)}$$

The only ball for which we do not have enough information is about B6, as it can be either less than H3 or greater than H3 and less than H4.

Combining (3), (4) and (5), we get,

$$B2 < H2 < B4 < H3 < B1 < H4 < B5 < H1 < B3$$

The sphere B6 can be placed in 2 positions, giving us two possible inequalities, which are

$$B2 < H2 < (B4, B6) < H3 < B1 < H4 < B5 < H1 < B3$$

$$B2 < H2 < B4 < H3 < (B1, B6) < H4 < B5 < H1 < B3$$

The position of B6 is unclear, and except for that, all the other positions are fixed.

In the first case, the number of pings possible can be calculated as 4 for B2, 3 for B4, 3 for B6, 2 for B1, 1 for B5 and 0 for B3. In total, the number of pings possible in the 1st case is $4 + 3 + 3 + 2 + 1 = 13$.

In the second case, the number of pings possible can be calculated as 4 for B2, 3 for B4, 2 for B6, 2 for B1, 1 for B5 and 0 for B3. In total, the number of pings possible in the 2nd case is $4 + 3 + 2 + 2 + 1 = 12$.

So, the total number of pings can be 12 or 13.

Hence, the correct answer is Option A.

VIDEO SOLUTION

38. 60

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{5}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b	1	3b/4		9b/16	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	
F	f	6	2f	6	3f/2	6

SI of E in 2016 is 60.

Hence, the correct answer is 60.

39.40

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b	1	3b/4		9b/16	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	
F	f	6	2f	6	3f/2	6

We know that F is ranked 6 and B is ranked 1 in 2016, and we are also given that the range of SI in 2016 and 2024 is 60. This means that the value of b-f = 60, and we can conclude that b is definitely greater than 60, as we know that f is an integer greater than 0.

We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

CASE 1: $b = 64$

If $b = 64$, then the value of f becomes 4, as we know $b - f = 60$.

If $f = 4$, then the value of F in 2024 is $3f/2 = 12/2 = 6$.

We already know the value of E in 2024 is 90.

In that case, the range of SI in 2024 is at least $90 - 6 = 84$, but we are given that the range is 60 in 2024, which is not possible in this case.

So, we can eliminate this case.

CASE 2: $b = 96$

If $b = 96$, then the value of f becomes 36, as we know $b - f = 60$.

If $f = 36$, then the value of F in 2024 is $3f/2 = 108/2 = 54$.

The value of B in 2024 = $9b/16 = (9 \times 96)/16 = 54$

We calculated the value of B and F to be 54 in 2024, which is not possible, as we are given that F had a lower SI than any other country in 2024, and we obtained its value to be the same as B .

So, we can eliminate this case.

CASE 3: $b = 80$

If $b = 80$, then the value of f becomes 20, as we know $b - f = 60$.

If $f = 20$, then the value of F in 2024 is $3f/2 = 60/2 = 30$.

The value of B in 2024 = $9b/16 = (9 \times 80)/16 = 45$

We can see that no condition is violated in this case, as F is less than B in 2024, and the range in 2024 from the obtained values as of now is $90 - 30 = 60$.

So, we can conclude that the value of $b = 80$ and $f = 20$.

Substituting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	$b = 80$	1	$3b/4 = 60$		$9b/16 = 45$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	$e = 60$	3	$5e/4 = 75$		$3e/2 = 90$	1
F	$f = 20$	6	$2f = 40$	6	$3f/2 = 30$	6

So, SI of F in 2020 is 40.

Hence, the correct answer is 40.

 VIDEO SOLUTION

40.84

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{25}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b	1	3b/4		9b/16	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	
F	f	6	2f	6	3f/2	6

We know that F is ranked 6 and B is ranked 1 in 2016, and we are also given that the range of SI in 2016 and 2024 is 60. This means that the value of b-f = 60, and we can conclude that b is definitely greater than 60, as we know that f is an integer greater than 0.

We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

CASE 1: b = 64

If b = 64, then the value of f becomes 4, as we know b - f = 60.

If f = 4, then the value of F in 2024 is $3f/2 = 12/2 = 6$.

We already know the value of E in 2024 is 90.

In that case, the range of SI in 2024 is at least $90 - 6 = 84$, but we are given that the range is 60 in 2024, which is not possible in this case.

So, we can eliminate this case.

CASE 2: b = 96

If b = 96, then the value of f becomes 36, as we know b - f = 60.

If f = 36, then the value of F in 2024 is $3f/2 = 108/2 = 54$.

The value of B in 2024 = $9b/16 = (9 \times 96)/16 = 54$

We calculated the value of B and F to be 54 in 2024, which is not possible, as we are given that F had a lower SI than any other country in 2024, and we obtained its value to be the same as B.

So, we can eliminate this case.

CASE 3: b = 80

If b = 80, then the value of f becomes 20, as we know b - f = 60.

If f = 20, then the value of F in 2024 is $3f/2 = 60/2 = 30$.

The value of B in 2024 = $9b/16 = (9 \times 80)/16 = 45$

We can see that no condition is violated in this case, as F is less than B in 2024, and the range in 2024 from the obtained values as of now is $90 - 30 = 60$.

So, we can conclude that the value of b = 80 and f = 20.

Substituting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	$5a/4$		$15a/8$	
B	b = 80	1	$3b/4 = 60$		$9b/16 = 45$	
C	c	2	$4c/5$		$28c/25$	
D	d	5	$2d$		$12d/5$	
E	e = 60	3	$5e/4 = 75$		$3e/2 = 90$	1
F	f = 20	6	$2f = 40$	6	$3f/2 = 30$	6

We know that C is ranked 2nd in 2016, and we know the values of 1st and 3rd to be 80 and 60. We also know that SI of C has to be integers in 2016, 2020 and 2024. So, for the value of SI of C to be an integer in 2024, the value of c has to be a multiple of 25.

So, we know that c lies between 80 and 60, and is also a multiple of 25. The only multiple of 25 that lies between 60 and 80 is 75. So, the value of c has to be 75, and the value of C in 2024 can be calculated as,

$$C \text{ in } 2024 = 28c/25 = (28 * 75)/25 = 84.$$

Hence, the correct answer is 84.

[VIDEO SOLUTION](#)



41. D

Let us assume the values of A, B, C, D, E and F to be a, b, c, d, e and f in 2016.

For A:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)a = \frac{5}{4}a$$

$$\text{Value in 2024} = \text{increased by 50\% from 2020} = \left(1 + \frac{50}{100}\right) \times \frac{5a}{4} = \frac{15}{8}a$$

For B:

$$\text{Value in 2020} = \text{decreased by 25\% from 2016} = \left(1 - \frac{25}{100}\right)b = \frac{3}{4}b$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times \frac{3b}{4} = \frac{9}{16}b$$

For C:

$$\text{Value in 2020} = \text{decreased by 20\% from 2016} = \left(1 - \frac{20}{100}\right)c = \frac{4}{5}c$$

$$\text{Value in 2024} = \text{increased by 40\% from 2020} = \left(1 + \frac{40}{100}\right) \times \frac{4c}{5} = \frac{28}{5}c$$

For D:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)d = 2d$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times 2d = \frac{12}{5}d$$

For E:

$$\text{Value in 2020} = \text{increased by 25\% from 2016} = \left(1 + \frac{25}{100}\right)e = \frac{5}{4}e$$

$$\text{Value in 2024} = \text{increased by 20\% from 2020} = \left(1 + \frac{20}{100}\right) \times \frac{5e}{4} = \frac{3}{2}e$$

For F:

$$\text{Value in 2020} = \text{increased by 100\% from 2016} = \left(1 + \frac{100}{100}\right)f = 2f$$

$$\text{Value in 2024} = \text{decreased by 25\% from 2020} = \left(1 - \frac{25}{100}\right) \times 2f = \frac{3}{2}f$$

We are given that the value of E is 90 in 2024. We can calculate the value of e as,

$$\frac{3e}{2} = 90$$

$$e = 60$$

The value of E in 2016 is 60, and its value in 2020 can be calculated as,

$$\frac{5e}{4} = \frac{5}{4} \times 60 = 75$$

We are given that B, C, E, and A had ranks 1, 2, 3, and 4, respectively, in 2016, and F has the lowest rank(6) in all the years.

Putting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b	1	3b/4		9b/16	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	
F	f	6	2f	6	3f/2	6

We know that F is ranked 6 and B is ranked 1 in 2016, and we are also given that the range of SI in 2016 and 2024 is 60. This means that the value of $b-f = 60$, and we can conclude that b is definitely greater than 60, as we know that f is an integer greater than 0.

We know that all the SI values in all the years are integers, and for SI of B to be an integer in 2024, the value of b must be a multiple of 16. The multiples of 16 that are greater than 60 and less than 100 are 64, 80 and 96.

CASE 1: $b = 64$

If $b = 64$, then the value of f becomes 4, as we know $b - f = 60$.

If $f = 4$, then the value of F in 2024 is $3f/2 = 12/2 = 6$.

We already know the value of E in 2024 is 90.

In that case, the range of SI in 2024 is at least $90 - 6 = 84$, but we are given that the range is 60 in 2024, which is not possible in this case.

So, we can eliminate this case.

CASE 2: $b = 96$

If $b = 96$, then the value of f becomes 36, as we know $b - f = 60$.

If $f = 36$, then the value of F in 2024 is $3f/2 = 108/2 = 54$.

The value of B in 2024 = $9b/16 = (9 * 96)/16 = 54$

We calculated the value of B and F to be 54 in 2024, which is not possible, as we are given that F had a lower SI than any other country in 2024, and we obtained its value to be the same as B.

So, we can eliminate this case.

CASE 3: $b = 80$

If $b = 80$, then the value of f becomes 20, as we know $b - f = 60$.

If $f = 20$, then the value of F in 2024 is $3f/2 = 60/2 = 30$.

The value of B in 2024 = $9b/16 = (9 * 80)/16 = 45$

We can see that no condition is violated in this case, as F is less than B in 2024, and the range in 2024 from the obtained values as of now is $90 - 30 = 60$.

So, we can conclude that the value of $b = 80$ and $f = 20$.

Substituting the values in the table, we get,

	2016	RANK in 2016	2020	RANK in 2020	2024	RANK in 2024
A	a	4	5a/4		15a/8	
B	b = 80	1	3b/4 = 60		9b/16 = 45	
C	c	2	4c/5		28c/25	
D	d	5	2d		12d/5	
E	e = 60	3	5e/4 = 75		3e/2 = 90	1
F	f = 20	6	2f = 40	6	3f/2 = 30	6

So, SI of B in 2024 is 45.

Hence, the correct answer is option D.

42.45

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50		12.5
Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humbleset is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humbleset.

CASE 2: Quackford

If the other city of Humbleset is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humbleset, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humbleset.

CASE 3: Zingaloo

If the other city of Humbleset is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humbleset, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humbleset, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humbleset, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

We can now assign the cities and the NURs the states and the PM values uniquely based on our calculations. After assigning the table looks like,

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50	Foggia	12.5
Splutterville	60	Whimshire	15
Quackford	70	Foggia	17.5
Mumpypore	80	Whimshire	20
Zingaloo	90	Humbleset	22.5
NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We calculated the PI of Whimshire to be 45.

Hence, the correct answer is 45.

VIDEO SOLUTION

43.35

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50		12.5
Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humbleset is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humbleset.

CASE 2: Quackford

If the other city of Humbleset is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humbleset, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humbleset.

CASE 3: Zingaloo

If the other city of Humbleset is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humbleset, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humbleset, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humbleset, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

We can now assign the cities and the NURs the states and the PM values uniquely based on our calculations. After assigning the table looks like,

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50	Foggia	12.5
Splutterville	60	Whimshire	15
Quackford	70	Foggia	17.5
Mumpypore	80	Whimshire	20
Zingaloo	90	Humbleset	22.5
NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We calculated the PI of Foggia to be 35.

Hence, the correct answer is 35.

 VIDEO SOLUTION

44. 50

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50		12.5
Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humbleset is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humbleset.

CASE 2: Quackford

If the other city of Humbleset is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humbleset, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humbleset.

CASE 3: Zingaloo

If the other city of Humbleset is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humbleset, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humbleset, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 10 = 45$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humbleset, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

We can now assign the cities and the NURs, the states, and the PM values uniquely based on our calculations. After assigning the table looks like,

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50	Foggia	12.5
Splutterville	60	Whimshire	15
Quackford	70	Foggia	17.5
Mumpypore	80	Whimshire	20
Zingaloo	90	Humbleset	22.5
NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We calculated the PI of Humbleset to be 50.

Hence, the correct answer is 50.

VIDEO SOLUTION

45. D

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humbleset.

So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humbleset has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humbleset.

We can also conclude that the PMs of Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo are 50, 60, 70, 80, and 90, respectively.

We are given that in the PI calculation, cities have a contribution of 25% and NUR has a contribution of 50%. So, we can calculate the contribution of each city in the calculation of PI by dividing their PI value by 4, and the contribution of NUR can be calculated by dividing the PM value by 2.

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humbleset	7.5
Noodleton	50		12.5
Splutterville	60		15
Quackford	70		17.5
Mumpypore	80		20
Zingaloo	90		22.5
NUR(Whimshire)	20/10	Whimshire	10/5
NUR(Foggia)	10/20	Foggia	5/10
NUR(Humbleset)	40	Humbleset	20

We are given that all the PI values are unique integers, and Humbleset has the highest PI value, while Foggia has the lowest.

For the PI value to be an integer, the cities with decimal contributions must be paired with another city with decimal contributions. Since all the decimals are 0.5, when two cities with 0.5 are added, the sum becomes 1, making the PI an integer.

The cities Splutterville and Mumpypore must be from the same state; both have integral contributions towards the PI, and if they are added to a decimal, then the result will be a decimal, which should not be the case. These two cities have to belong to either the state of Whimshire or Foggia, but their combined contribution towards PI is $15 + 20 = 35$.

We already know that for the state of Humbleset, the PI is highest, and the two contributions towards the PI are 7.5 and 20, totalling 27.5.

There are three possibilities for the other city of Humbleset, which are Noodleton, Quackford and Zingaloo.

CASE 1: Noodleton

If the other city of Humbleset is Noodleton, then its PI would be $27.5 + 12.5 = 40$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humbleset.

CASE 2: Quackford

If the other city of Humblest is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humblest, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humblest.

CASE 3: Zingaloo

If the other city of Humblest is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

If the PI of the other state is 40,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 10 = 40$.

We can see that PI of the other two states is 40, which violates one of the conditions given.

If the PI of the other state is 45,

Then, for Humblest, the PI is $7.5 + 22.5 + 20 = 50$.

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The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 17.5 + 5 = 35$.

We can see that all the conditions are satisfied in this case, and we can conclude that Zingaloo is the second city of Humblest, and also we know that PI is the least for Foggia. So, the PI of Foggia is 35, and the PI of Whimshire is 45.

We can now assign the cities and the NURs, the states, and the PM values uniquely based on our calculations. After assigning the table looks like,

CITY and NUR	PM	STATE	Contribution in PI
Blusterburg	30	Humblest	7.5
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NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humblest)	40	Humblest	20

We can see that Noodleton and Quackford belong to the same state among the options.

Hence, the correct answer is option C.

[VIDEO SOLUTION](#)



46.9

We are given that the PMs of the six cities, Blusterburg, Noodleton, Splutterville, Quackford, Mumpypore, and Zingaloo, are in increasing order.

We are also told there exists a single pair of NUR and city such that the PM of NUR is greater than the city and the NUR, city of the pair belongs to Humblest. So, we can conclude that PMs of NURs of Whimshire and Foggia are 10 and 20 in some order.

We also know that out of all the cities, Blusterburg has the lowest PM. So, the PM of Blusterburg has to be 30, and the PM of NUR of Humblest has to be 40 for the above condition to be satisfied, and we can also conclude that Blusterburg is a city of Humblest.

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However, we are told that the PI values are unique, and Humblest has the highest PI value; in either case, this condition is not satisfied.

So, we can eliminate the case of Noodleton being the second city of Humbleset.

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If the other city of Humbleset is Quackford, then its PI would be $27.5 + 17.5 = 45$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

However, we are told that the PI values are unique, and Humblest has the highest PI value. So, we can eliminate the case of PI of NUR being 10 for the other state.

So, for Humbleset, the PI is $7.5 + 17.5 + 20 = 45$.

For one of the other 2 states, the PI is $15 + 20 + 5 = 40$

The PI of the other state can be calculated by adding all the leftover PIs, which is $12.5 + 22.5 + 10 = 45$.

We can see that PI of Humbleton and one of the other two states is 45, which violates one of the conditions given.

So, we can eliminate the case of Quackford being the second city of Humbleset.

CASE 3: Zingaloo

If the other city of Humbleset is Zingaloo, then its PI would be $27.5 + 22.5 = 50$.

We already obtained the sum of the PIs of one of the other 2 states to be 35 without including the PI of NUR. We know that the NUR contribution for that city would be either 5 or 10. If it is 5, then the PI of that state would be 40, and if it is 10, then the PI of that state would be 45.

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We can now assign the cities and the NURs, the states, and the PM values uniquely based on our calculations. After assigning the table looks like,

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NUR(Whimshire)	20	Whimshire	10
NUR(Foggia)	10	Foggia	5
NUR(Humbleset)	40	Humbleset	20

We can see that for all 6 cities and 3 NURs, the PMs and the city to which they belong can be identified uniquely. So, in total, we can identify the values uniquely for all 9 of them.

Hence, the correct answer is 9.

[VIDEO SOLUTION](#)

Quant

47. B

Let's assume that $x^2 + 2x - 3 = t$

$9^{x^2+2x-3} - 4(3^{x^2+2x-2}) + 27 = 0$ can be written as $9^t - 4(3^{t+1}) + 27 = 0$

$3^{2t} - 12(3^t) + 27 = 0$

Let's assume that $3^t = y$

$$y^2 - 12y + 27 = 0$$

$$(y - 9)(y - 3) = 0$$

$$y = 3 \text{ or } 9 \Rightarrow t = 1 \text{ or } 2$$

Let's solve when $t = 1$

$$x^2 + 2x - 3 = 1 \Rightarrow x^2 + 2x - 4 = 0$$

$$b^2 - 4ac = 4 + 16 = 20$$

Positive, so the equation has real roots.

Product of possible value of $x = -4$

Let's solve for $t = 2$

$$x^2 + 2x - 3 = 2 \Rightarrow x^2 + 2x - 5 = 0$$

$$b^2 - 4ac = 4 + 20 = 24$$

Positive, so the equation has real roots.

Product of possible value of $x = -5$

The product of all values = 20

[VIDEO SOLUTION](#)

48.49

Let the copies sold on days 1 through 9 be $d_1, \dots, d_9$

From the averages:

$$d_1 + \dots + d_7 = 7 \times 60 = 420$$

$$d_1 + \dots + d_8 = 8 \times 63 = 504$$

$$\text{So } d_8 = 504 - 420 = 84. \text{ Then } d_9 = 84 - 11 = 73$$

The sum from day 2 to day 9 is $8 \times 66 = 528$

$$d_2 + \dots + d_9 = (d_1 + \dots + d_8) - d_1 + d_9 = 504 - d_1 + 73 = 577 - d_1$$

$$\text{Thus } 577 - d_1 = 528 \Rightarrow d_1 = 49$$

[VIDEO SOLUTION](#)

49.A

We are asked to solve

$$x^2 - |x + 9| + x > 0$$

Split into two cases based on the absolute value.

$$\text{Case 1: } x + 9 \geq 0 \Rightarrow x \geq -9$$

$$|x + 9| = x + 9$$

$$\text{Inequality becomes: } x^2 - (x + 9) + x > 0 \Rightarrow x^2 - 9 > 0 \Rightarrow (x - 3)(x + 3) > 0$$

$$\text{So } x < -3 \text{ or } x > 3. \text{ Combined with } x \geq -9, \text{ we get } x \in [-9, -3) \cup (3, \infty)$$

$$\text{Case 2: } x + 9 < 0 \Rightarrow x < -9$$

$$|x + 9| = -(x + 9) = -x - 9$$

$$\text{Inequality becomes: } x^2 - (-x - 9) + x > 0 \Rightarrow x^2 + 2x + 9 > 0$$

The quadratic $x^2 + 2x + 9$ has discriminant $(4 - 36 = -32 < 0)$, so always positive. But in this case ($x < -9$), so inequality is satisfied. Thus ($x < -9$) also works.

$$x < -3 \text{ or } x > 3$$

$$\text{So the solution set is } (-\infty, -3) \cup (3, \infty)$$

[VIDEO SOLUTION](#)

50.C

Let the marked price be M and the initial discount rate be d . The cost price is 1650. A profit of 20% means the selling price is

$$SP = 1650 \times 1.2 = 1980$$

With discount d ,

$$M(1 - d) = 1980$$

If the discount is doubled, the selling price becomes $(M(1 - 2d))$, and the profit is 110, so

$$M(1 - 2d) - 1650 = 110 \Rightarrow M(1 - 2d) = 1760$$

Subtracting the two equations:

$$M(1 - d) - M(1 - 2d) = 1980 - 1760 \Rightarrow Md = 220 \Rightarrow M = \frac{220}{d}$$

Plug into the first equation:

$$\frac{220}{d}(1 - d) = 1980 \Rightarrow \frac{1 - d}{d} = 9 \Rightarrow 1 - d = 9d \Rightarrow d = 0.1$$

So the initial discount rate is 10% and the marked price is

$$M = \frac{220}{0.1} = 2200$$

Now, let the discount rate be (r) such that the profit percentage equals the discount percentage. Then

$$\frac{2200(1-r)-1650}{1650} = r$$

Simplify:

$$2200(1-r) - 1650 = 1650r \Rightarrow 2200 - 2200r - 1650 = 1650r \Rightarrow 550 = 3850r \Rightarrow r = \frac{550}{3850} \approx 0.142857$$

Thus, the required discount rate is approximately 14%

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51.17

Sure! We can solve it without explicitly introducing new variables for substitution.

We are given $(m+2n)(2m+n) = 27$ and we want to maximize $2m-3n$

Because m and n are integers, both $(m+2n)$ and $(2m+n)$ are going to be integers.

Factor pairs of 27 (including negatives) are: $(1, 27), (3, 9), (9, 3), (27, 1), (-1, -27), (-3, -9), (-9, -3), (-27, -1)$

For each factor pair (a, b) , take $a = m+2n, b = 2m+n$ and solve for integers (m, n) .

If we notice $a+b = (m+2n) + (2m+n) = 3(m+n)$

So, basically the sum of the two numbers has to be a multiple of 3.

So, the ordered pairs we will consider are $(3, 9), (9, 3), (-3, -9), (-9, -3)$

For $(a, b) = (9, 3)$:

$$m+2n=9, \quad 2m+n=3$$

Upon solving, we get $n=5$ and $m=-1$

For $(a, b) = (3, 9)$:

$$m+2n=3, \quad 2m+n=9$$

Upon solving, we get $n=-1$ and $m=5$

$$\text{Then } 2m-3n = 2 \cdot 5 - 3(-1) = 10 + 3 = 13$$

For negative factor pairs, we similarly get integer solutions $(-5, 1)$ and $(1, -5)$, giving $2m-3n = -13$ and 17, respectively.

Thus, the maximum value of $(2m-3n)$ among all solutions is 17

[VIDEO SOLUTION](#)

52.25

$$625 = 5^4 \text{ and } 128 = 2^7$$

$$\text{So, } (625)^{65}, (128)^{36} = 5^{260}, 2^{252} = (2^{252} 5^{252}) \cdot 5^8 = 10^{252} \cdot 5^8$$

Thus, the number equals 5^8 followed by 252 zeros. Now $(5^8=390625)$, and the sum of the digits of (390625) is

$$3+9+0+6+2+5=25.$$

Zeros add nothing, so the required sum of digits is 25.

[VIDEO SOLUTION](#)

53.A

Let's assume that the common root is r .

The sum of the roots of the first equation is $5/3$ and that of the second equation is 1.

We want the sum of the other two roots:

$$\text{Sum} = \left(\frac{5}{3} - r\right) + (1 - r) = \frac{8}{3} - 2r$$

We now need to express r in terms of p and q .

Since r is a common root, it satisfies:

$$3r^2 - 5r + p = 0 \quad (1)$$

$$2r^2 - 2r + q = 0 \quad (2)$$

Eliminate r^2 .

Multiply (2) by 3:

$$6r^2 - 6r + 3q = 0$$

Multiply (1) by 2:

$$6r^2 - 10r + 2p = 0$$

Subtract:

$$(6r^2 - 6r + 3q) - (6r^2 - 10r + 2p) = 0$$

$$4r + 3q - 2p = 0$$

$$r = \frac{2p-3q}{4}$$

Now substitute into $\frac{8}{3} - 2r$:

$$\frac{8}{3} - 2 \left(\frac{2p-3q}{4} \right)$$

$$= \frac{8}{3} - \frac{2p-3q}{2}$$

$$= \frac{8}{3} - p + \frac{3}{2}q$$

VIDEO SOLUTION

54. A

$$64 = 8^2 \text{ and } 512 = 8^3$$

$$\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512} (\sqrt{y}z) = 4,$$

$$\text{Property of log: } \log_{b^m} a^n = \frac{n}{m} \log_b a$$

$$\log_{8^2} x^2 + \log_8 \sqrt{y} + 3 \log_{8^3} (\sqrt{y}z) = 4$$

Using the above-mentioned property, the expression becomes $\log_8 x + \log_8 \sqrt{y} + \log_8 (\sqrt{y}z) = 4$

$$\log_8 x \sqrt{y} \cdot (\sqrt{y}z) = 4$$

$$\log_8 xyz = 4$$

$$xyz = 8^4 = 2^{12}$$

Using AM-GM inequality

$$\frac{(x+y+z)}{3} \geq \sqrt[3]{xyz}$$

$$\frac{(x+y+z)}{3} \geq 2^4$$

$$(x+y+z) \geq 48$$

VIDEO SOLUTION

55. 8

Let the river speed be v km/h. For Sneha still-water speed = 6 km/h:

Upstream speed = $(6-v)$, downstream speed = $(6+v)$.

$$\text{Given } \frac{14}{(6-v)} - \frac{14}{(6+v)} = 48 \text{ minutes} = 0.8 \text{ hours.}$$

$$\text{So, } \frac{14(6+v-6+v)}{36-v^2} = 0.8 \Rightarrow \frac{28v}{36-v^2} = 0.8$$

$$28v = 0.8(36-v^2) = 28.8 - 0.8v^2 \Rightarrow 0.8v^2 + 28v - 28.8 = 0$$

$$\text{Multiply by 5: } 4v^2 + 140v - 144 = 0$$

$$v^2 + 35v - 36 = 0$$

We get $v = 1$

Thus, the river speed is 1 km/h. For Rita, the still water speed is 5 km/h:

Upstream speed = $5-1=4$ km/h, Downstream Speed = $(5+1=6)$ km/h.

If she rows d km upstream and returns d km downstream, the total time

$$\frac{d}{4} + \frac{d}{6} = d \left(\frac{1}{4} + \frac{1}{6} \right) = d \cdot \frac{5}{12} \text{ hours}$$

$$\text{This equals 100 minutes} = 100/60 = \frac{5}{3} \text{ hours.}$$

$$\text{So, } d \cdot \frac{5}{12} = \frac{5}{3} \Rightarrow d = 4 \text{ km}$$

Total distance covered $2d=8$ km.

VIDEO SOLUTION

100+ CAT DILR Questions

cracku



56. 12

Our task is to minimise $3a + 2b + c$.

Here, the coefficient for c is the minimum.

$$3ac = 8(a+b)$$

We know that a , b , and c are natural numbers. So, the product ac should definitely be a multiple of 8.

$$\text{Case 1: } a = 1, c = 8 \text{ and } b = 2 \Rightarrow 3a+2b+c = 15$$

$$\text{Case 2: } a = 2, c = 4 \text{ and } b = 1 \Rightarrow 3a+2b+c = 12$$

So, 12 is the correct answer.

VIDEO SOLUTION

57. A

We check where $f(x) = \frac{x}{2x-1}$ or $g(x) = \frac{x}{x-1}$, or their compositions, become undefined.

First, $f(x)$ is undefined at $x = \frac{1}{2}$ and $g(x)$ is undefined at $x = 1$.

Next, for $f(g(x)) = \frac{x}{x+1}$

Since, the denominator can't be zero, $x = -1$ must also be excluded.

For $g(f(x)) = \frac{x}{1-x}$

$\Rightarrow x = 1$ is not possible, which is already excluded.

So the values at which $h(x) = f(g(x)) + g(f(x))$ is undefined are $-1, \frac{1}{2}, 1$

VIDEO SOLUTION

58. D

Let the annual interest rate be (r) (in decimal). Discount the two instalments to present value:

$$\frac{530}{1+r} + \frac{594}{(1+r)^2} = 1000$$

Set $x = \frac{1}{1+r}$. Then

$$594x^2 + 530x - 1000 = 0$$

$$\text{Discriminant} = 530^2 + 4 \cdot 594 \cdot 1000 = 280900 + 2376000 = 2656900 = 1630^2.$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-530 + 1630}{2 \cdot 594} = \frac{1100}{1188} = \frac{275}{297}$$

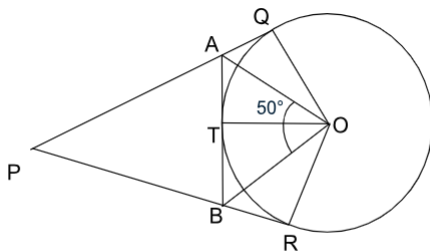
$$\text{So } 1+r = \frac{297}{275}$$

$$r = \frac{297 - 275}{275} = \frac{22}{275} = \frac{2}{25} = 0.08 = 8\%$$

VIDEO SOLUTION

59. 80

We can draw the following diagram as per the given question :



Now, taking $\triangle AOT$ and $\triangle AOQ$

AO is the common side

AT=AQ (Tangents drawn from an external point are equal in length)

OT=OQ (radius of circle)

So, by S.S.S., $\triangle AOT$ is congruent to $\triangle AOQ$

So, by C.P.C.T., $\angle AOT = \angle AOQ$

Similarly, for $\triangle BOT$ and $\triangle BOR$, we can do the same thing,

So, we can say $\triangle BOT$ and $\triangle BOR$ will also be congruent

$$\text{Now, } \angle QOR = \angle AOQ + \angle AOB + \angle ROB = \angle AOT + \angle AOB + \angle BOT = 2\angle AOB = 2 \times 50^\circ = 100^\circ$$

Now, in quadrilateral OQPR,

$$\angle OQP = \angle ORP = 90^\circ \text{ (Tangents drawn to a point is perpendicular to the radius drawn at the same point)}$$

Now, sum of angles of a quadrilateral is 360°

$$\text{So, } \angle APB + \angle QOR = 360^\circ - (90^\circ + 90^\circ) = 180^\circ$$

$$\text{So, } \angle APB = 180^\circ - 100^\circ = 80^\circ$$

VIDEO SOLUTION

60. D

The divisors of the given number will have the form $2^a * 3^b * 5^c * 7^d$ with $0 \leq a \leq 6, 0 \leq b \leq 5, 0 \leq c \leq 3, 0 \leq d \leq 2$

Because the divisors should be in the form $3r+1$, it cannot be divisible by 3, so $(b=0)$.

Reduce modulo 3: $2 \bmod 3 = 2, 5 \bmod 3 = 2, 7 \bmod 3 = 1$

Hence, $2^a 5^c 7^d \equiv 2^{a+c} \cdot 1^d \equiv 2^{a+c} \pmod{3}$

2^k will be in the form $3r+1$ only when k is even. So, we need $a+c$ to be even

$a \in 0, \dots, 6$ has 4 even, 3 odd values

$c \in 0, \dots, 3$ has 2 even, 2 odd

Number of (a, c) with $(a+c)$ even is $4 \cdot 2 + 3 \cdot 2 = 8 + 6 = 14$

For each such pair, there are 3 choices for $d = 0, 1, 2$. Thus, total divisors in the form $(3r+1)$ equals $14 \times 3 = 42$.

VIDEO SOLUTION

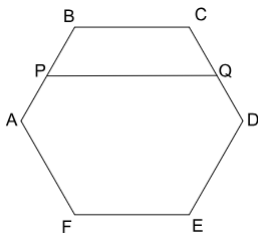
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61. B

Let the side of the hexagon be a .

The area of the whole hexagon is going to be $\frac{3\sqrt{3}a^2}{2}$



We know that the longest diagonal of a hexagon is 2 times the side of the hexagon.

$$PQ = \frac{(BC+AD)}{2} = \frac{(a+2a)}{2} = 1.5a$$

The area of the trapezium = Average of parallel sides * Height = $\frac{(BC+PQ)}{2} \cdot h = \frac{5}{4}a \cdot h$

The distance between BC and AD will be $\sqrt{3}a$, that is, the length of diagonal BF.

The distance between BC and AD will be $\frac{\sqrt{3}a}{2}$

And the height of the trapezium BCQP will be $\frac{\sqrt{3}a}{4}$

$$\text{Area of trapezium} = \frac{5}{4}a \cdot \frac{\sqrt{3}a}{4} = \frac{5\sqrt{3}a^2}{16}$$

$$\text{Area of hexagon ABCDEF} = \frac{3\sqrt{3}}{2}a^2$$

$$\text{We have to find the ratio} = \frac{5\sqrt{3}a^2}{16} : \frac{3\sqrt{3}}{2}a^2 = 5:24$$

VIDEO SOLUTION

62. 2

Given expression: $(a-b)^2 + (a-c)^2 + (a-d)^2$

The given expression has just the sum of squares of the terms. So, the minimum value is either zero or positive.

If we can make all the values equal, we can get zero. But since all the values are integers and the sum 46 is not divisible by 4, we can't make everything equal.

So, the nearest four values are 12, 11, 11, 12.

With this, the minimum value is $(12-11)^2 + (12-11)^2 + (12-12)^2 = 2$

VIDEO SOLUTION

63. A

Let Lakshmi's income = A and expenditure = B.

Let Meenakshi's income = C and expenditure = D.

From B:D = 2:3, take B = 2k and D = 3k.

$$\text{From A:D} = 6:7, A = \frac{6}{7}D = \frac{6}{7} \cdot 3k = \frac{18}{7}k.$$

$$\text{Lakshmi's saving} = A - B = \frac{18}{7}k - 2k = \frac{4}{7}k.$$

Let Meenakshi's saving = $C - D = C - 3k$. Given the savings ratio $\frac{\frac{4k}{7}}{C-3k} = \frac{4}{9}$.

$$\text{Solve: } \frac{4}{7}k \cdot \frac{9}{4} = C - 3k \Rightarrow \frac{9}{7}k = C - 3k \Rightarrow C = \frac{30}{7}k$$

$$\text{So incomes ratio } A : C = \frac{18}{7}k : \frac{30}{7}k = 18 : 30 = 3 : 5.$$

VIDEO SOLUTION

64. B

Let the first term be a and the common ratio be r .

Given

$$a + ar + ar^2 = 52 \quad \text{and} \quad a^2r + a^2r^2 + a^2r^3 = 624.$$

$$\text{From the first equation } a(1 + r + r^2) = 52, \text{ so } a = \frac{52}{1 + r + r^2}.$$

Substitute into the second:

$$\frac{52^2}{(1+r+r^2)^2} * (r + r^2 + r^3) = 624$$

$$\frac{r}{(1+r+r^2)} = \frac{3}{13}$$

$$3r^2 - 10r + 3 = 0$$

$$(3r - 1)(r - 3) = 0$$

Simplify to get an equation in (r) ; its real solutions are $r = \frac{1}{3}$ and $r = 3$. For an infinite geometric progression, we must have $(|r| < 1)$, so $r = \frac{1}{3}$.

Now

$$a = \frac{52}{1 + \frac{1}{3} + \frac{1}{9}} = \frac{52}{\frac{13}{9}} = 36$$

The sum of the infinite progression is

$$S = \frac{a}{1-r} = \frac{36}{1-\frac{1}{3}} = \frac{36}{\frac{2}{3}} = 54.$$

VIDEO SOLUTION

65. A

Let coffee price = C Rs/kg and cocoa price = K Rs/kg.

From the two given mixtures:

$$0.16C + 0.84K = 240$$

$$0.36C + 0.64K = 320$$

Multiply both equations by 100 to remove decimals:

$$16C + 84K = 24000$$

$$36C + 64K = 32000$$

Subtract the first from the second:

$$(36C + 64K) - (16C + 84K) = 32000 - 24000$$

$$20C - 20K = 8000 \implies C - K = 400$$

Put $C = K + 400$ into $16C + 84K = 24000$:

$$16(K + 400) + 84K = 24000$$

$$16K + 6400 + 84K = 24000$$

$$100K = 17600 \implies K = 176$$

So $C = 176 + 400 = 576$ (Rs/kg).

For the new mixture priced at Rs 376/kg, let the coffee fraction be p . Then

$$p \cdot 576 + (1 - p) \cdot 176 = 376$$

Upon solving $p = \frac{1}{2}$

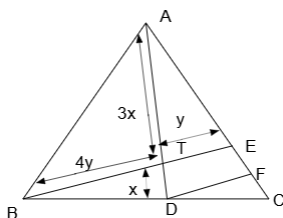
Thus, coffee is (50%) of the new mixture. In 10 kg of this mixture, coffee = $10 \times 0.5 = 5$ kg

VIDEO SOLUTION

CAT Expected Questions PDF



66. B



In the question, it is given $AD : AT = 4 : 3$

So, $TD : AT = 1 : 3$

Let, $TD = x$, $AT = 3x$

Also, $BE : BT = 5 : 4$

So, $BT : TE = 4 : 1$

Let, $BT = 4y$, $TE = y$

Given, DF is parallel to BE

So, DF is parallel to TE

So, $\triangle ATE \triangle ADF$ (by A.A., as $\angle ATE = \angle ADF$ and $\angle AET = \angle AFD$)

$$\text{So, } \frac{AT}{AD} = \frac{TE}{DF}$$

$$\text{So, } \frac{3x}{x + 3x} = \frac{y}{DF}$$

$$\text{So, } DF = \frac{4y}{3}$$

Also, since DF is parallel to BE,

So, $\triangle CFD \triangle CBE$ (by A.A., as $\angle CFD = \angle CEB$ and $\angle CDF = \angle CBE$)

$$\text{So, } \frac{CD}{BC} = \frac{DF}{BE}$$

$$\text{So, } \frac{CD}{BC} = \frac{\frac{4y}{3}}{5y} = \frac{4y}{15y} = \frac{4}{15}$$

$$\text{So, } \frac{CD}{BD} = \frac{4}{11}$$

So, $BD : CD = 11 : 4$

[VIDEO SOLUTION](#)

67. 340

Let Chandan's one-day work = x .

Then Bipin = $2x$ and Ankita = $4x$.

All three work together for the first 20 days. Their combined daily work = $x + 2x + 4x = 7x$.

Work done in 20 days = $20 \times 7x = 140x$.

Bipin leaves after 20 days; Ankita and Chandan continue for the remaining (60-20=40) days.

Their combined daily work = $4x + x = 5x$. Work done in 40 days = $40 \times 5x = 200x$.

Total work = $140x + 200x = 340x$.

We know that Chandan does x work in a day. So, the number of days it takes him to finish the work is 340.

Answer: 340 days.

[VIDEO SOLUTION](#)

68. C

Let the total amount be 100 units (taking 100 makes percentage calculations easy).

Pinu receives 20% of the total, so Pinu gets 20 units.

The amount left after giving Pinu his share is $100 - 20 = 80$ units.

Meena receives 40% of this remaining amount, so Meena gets 40% of 80 = 32 units.

Now, the amount left for Rinu and Seema together is $80 - 32 = 48$ units.

Seema receives 20% less than Pinu. Since Pinu gets 20 units, Seema gets

$20 - 20\% \text{ of } 20 = 20 - 4 = 16$ units.

Out of the 48 units left for Rinu and Seema, Seema's share is 16 units, so Rinu gets

$48 - 16 = 32$ units.

Therefore, Pinu receives 20 units, and Rinu gets 32 units.

The required ratio of the amounts received by Pinu : Rinu = $20 : 32 = 5 : 8$.

[VIDEO SOLUTION](#)